

SUMANT BAGRI

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EXPERIENCE

Ocado Technology

Toronto, Canada

Software Engineer II, Product and Applied ML | C++, Go, Python, CUDA, ROS2, gRPC, Triton

May 2023 – Present

Shipped production ML perception, robotics control, and warehouse automation software across the full stack.

- Led the architectural design of a 3D perception pipeline using the Kedro atomic-node framework, integrating fine-tuned Mask2Former models, packaged as a dockerized gRPC microservice, achieving a 30% improvement in mAP.
- Engineered a backend-agnostic inference gateway unifying Triton and TorchServe, with dynamic batching and GPU memory pooling across multi-model workloads, reducing system memory overhead by 30%+.
- Designed and implemented scalable, high-performance, robotic grasping and placement algorithms in C++ and Python, improving item handling throughput by 63% with a 92%, fully autonomous, production cycle rate.
- Architected a real-time Linux teleoperation stack in ROS2/DDS with zero-copy shared-memory transport, streamlining kinematic demonstration collection for imitation learning policies and 6-DoF IMU calibration.
- Built a remote power-cycle system for the SORT production fleet in Go, eliminating over 9,000 hr/week of manual, on-site intervention.
- Prototyped a novel palletization algorithm achieving over 90% autonomous packing efficiency, showcased at ProMat 2025.
- Engineered a GPU-accelerated collision detector for 6-DoF manipulators using PhysX, benchmarked against FCL and deployed into Kindred AI's production stack, achieving 4x GPU speedup and 65–70x CPU speedup.

Flow Traders

Hong Kong

Trading Operations Engineer | Python, Ansible, Airflow, Kubernetes, Bash, Linux

Oct 2020 – Apr 2022

Deployed high-frequency, production trading infrastructure, maintaining exchange connectivity across 20+ global venues.

- Led the design of on-premise Apache Airflow infrastructure on Kubernetes and developed 15+ DAG pipelines covering Bloomberg price ingestion, futures rollover, and end-of-day trade analysis.
- Achieved sub-microsecond deterministic execution via kernel-bypass networking while reducing hardware footprint by approximately 50% through NUMA-aware process pinning on bare-metal servers.

The Walt Disney Company

Mumbai, India

Applied ML Engineer Intern | Python, REST, BigQuery

May 2018 – Jul 2018

Designed ML-driven behavioral analytics, predicting user engagement for Disney Interactive products.

- Built an automated BigQuery data pipeline consolidating fragmented sales and user data (+33% processing efficiency), feeding a PCA-reduced gradient-boosted regressor predicting in-game click-through rates and session duration ($R^2 = 0.83$).

IBM

Mumbai, India

Cognitive Cloud Application Developer Intern | Java, IBM Bluemix, Cloud Foundry

Nov 2016 – Jan 2017

Built cloud-native AI applications on IBM's managed PaaS infrastructure.

- Developed cloud-native Java applications on IBM Bluemix via Cloud Foundry DevOps pipelines, integrating Watson APIs (Personality Insights, Tradeoff Analytics) for AI-driven behavioral profiling and decision optimization.

EDUCATION

University of Toronto

Toronto, Canada

M.Sc in Applied Computing, Computer Science; AI Concentration — GPA: 4.0 / 4.0

Sep 2022 – Dec 2023

IIT Bombay

Mumbai, India

B.Tech & M.Tech, Mechanical Engineering; Minor in Electrical Engineering — GPA: 8.6 / 10.0

Aug 2015 – Aug 2020

TECHNICAL SKILLS

Languages: Python, C++20, Go, Java, CUDA, Bash

ML & AI: PyTorch, TensorRT, ONNX, Triton Server, TorchScript, vLLM, HuggingFace Transformers, LangChain, scikit-learn

Robotics & Vision: ROS2, MoveIt2, Open3D, OpenCV, OpenMP, gRPC, Modbus TCP/IP

Infra & Tools: Docker, Kubernetes, Nomad, Apache Airflow, Kedro, Git, Pytest

PROJECTS

LLM Inference Benchmark Dashboard | Python, vLLM, HuggingFace, llama.cpp, LangChain, Plotly Dash

- Developed an inference benchmarking suite for vLLM, HuggingFace, and llama.cpp unified under a common LangChain interface.
- Benchmarked Mistral-7B across 4 quantization levels with p99 latency, tokens/sec, and peak VRAM metrics persisted to SQLite.
- Built a live Plotly Dash dashboard visualizing TTFT distributions, throughput scatter, and scaling curves.

Diffusion Models on Edge | PyTorch, TensorRT, ONNX Runtime, TorchScript

- Implemented LDM and Stable Diffusion v1.4 inference pipelines in PyTorch, covering DDIM sampling, UNet2D, and VAE decoding.
- Benchmarked FP16, TorchScript JIT, ONNX, and TensorRT across RTX 3070 and Jetson Nano, achieving a 4.7x latency reduction.

PUBLICATIONS

Bagri, S., et al. "Tool wear and remaining useful life prediction in micro-milling." *J. Manufacturing Processes* (2024)

Manwar, A., et al. "Online tool condition monitoring in micromilling using LSTM." *J. Intell. Manuf.* (2025)